

CADEMAS-ML: A Software Tool for Context-Aware Cooperative and Explainable Automated Decision-Making

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ABSTRACT

Complex organizational decisions often require several actors to combine their assessments while accounting for the surrounding operational and strategic context. The Cooperative Automated Decision-Making Systems (CADEMAS) framework was recently proposed to represent this form of cooperative and context-aware decision-making. Previous CADEMAs instantiations, however, cover only specific configurations of a framework that admits substantially broader combinations of automated decision components, contextual models, and aggregation mechanisms. This flexibility is valuable in practice, but it also makes the framework difficult to operationalize without dedicated software support. This paper presents CADEMAs-ML, a software tool that implements a configurable class of CADEMAs systems in which automated decision-making components are machine-learning models and contextual knowledge is represented through fuzzy logic. The tool integrates and weights multiple predictive models, supports configurable fuzzy contexts and alternative aggregation operators, and makes the trade-off between predictive risk and contextual suitability explicit. It also incorporates explainability and robustness analysis to help users inspect and assess the resulting decisions. An employee-attribution case study illustrates how the same predictive evidence can lead to different prioritizations under alternative contextual scenarios.

KEYWORDS

automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization; cooperative systems

1. Introduction

Decision-making in real-world organizations is rarely a matter of applying a single analytical model to a well-defined set of inputs. Complex decisions often involve multiple

stakeholders, organizational units, or analytical components that contribute different pieces of evidence to the same problem. These contributions may reflect different objectives, information sources, areas of expertise, or levels of reliability, and therefore need to be coordinated before they can support a coherent course of action.

This perspective becomes particularly important when automated decision-making is used to support prediction, classification, or risk assessment. An automated decision-making (ADM) component can produce a probability, rating, recommendation, or risk score, but that output alone does not determine how the result should be acted upon. The appropriate course of action may also depend on organizational priorities, available resources, departmental objectives, regulatory constraints, or the strategic conditions surrounding the decision. In short, decisions are taken in a context, not in a vacuum.

Different ADM components may also provide complementary views of the same cases, while the final decision may need to reflect contextual knowledge provided by domain experts or organizational policies.

These considerations motivated the Cooperative Automated Decision-Making Systems (CADEMAS) framework proposed by Novoa-Hernández, Pelta, Godz, Verdegay, and Buendia-Carrillo (2026). CADEMÁS coordinates heterogeneous ADM components while incorporating the context in which their outputs are interpreted. Its central idea is to keep the generation and integration of decision evidence separate from the contextual evaluation of that evidence, so that a collective assessment can later be combined with organizational priorities, policies, or scenario-dependent considerations. Section 2.1 formalizes this architecture.

The framework builds on cooperative and multi-agent decision-support systems, in which heterogeneous components contribute to a common decision process (Bui & Lee, 1999; Vahidov & Fazlollahi, 2004), and on multiperson and group decision-support research, where heterogeneous information and preferences are traditionally combined through aggregation, negotiation, and consensus-seeking mechanisms (Herrera-Viedma, Herrera, & Chiclana, 2002; Salo, 1995). Similar principles extend naturally to automated settings, where the cooperating components are software-based decision units rather than human decision makers.

A particularly relevant scenario is prioritization under a limited intervention budget: given a set of entities, which ones should receive attention first? Examples include attrition-risk prevention Godz, Novoa-Hernández, and Pelta (2026), financial rescue for small and medium-sized companies Navarro-Galera, Lara-Rubio, Novoa-Hernández, and Cruz Corona (2025), and student-dropout prevention (BUSCAR REF).

In such problems, CADEMÁS combines the assessments of multiple ADMs with the suitability of each entity for a given decision context. For instance, under an economic-crisis policy one may prioritize companies with many employees, whereas under a growth policy one may favor those with high expected impact. This connection is developed formally in the full manuscript. Fuzzy rules offer a convenient way

to represent such contexts (Lamata, Pelta, & Verdegay, 2020; Pérez-Cañedo, Porras, Pelta, & Verdegay, 2023), combining quantitative information with expert knowledge and linguistic assessments.

Users may also need to understand why a particular entity receives a given prediction, especially when predictions contribute to decisions with organizational or human consequences (Barredo Arrieta et al., 2020). Local feature attributions can help inspect the factors associated with individual predictions (Lundberg et al., 2020; Lundberg & Lee, 2017).

In addition, the ordering of cases may change when the contextual definition, the aggregation mechanism, or the balance assigned to predictive and contextual information is revised, even if the underlying models remain unchanged. Robustness analysis reveals which priorities are stable and which cases are particularly sensitive to these policy choices (Barredo Arrieta et al., 2020).

To the best of our knowledge, no integrated tool currently supports this complete prioritization workflow.

This paper presents CADEMAs-ML to address that gap. The tool implements a configurable class of CADEMAs systems for prioritization problems, with machine-learning ADM units and fuzzy contextual knowledge. Beyond the core CADEMAs capabilities, it incorporates case-level explanation and robustness analysis as natural extensions of the workflow.

The remainder of this excerpt reviews the CADEMAs framework and related work on cooperative and fuzzy decision-support systems (Section 2).

2. Background and related works

This section sets out the conceptual foundations on which CADEMAs-ML builds. We first revisit the CADEMAs framework in its general form (Section 2.1). We then situate the tool among related cooperative-decision frameworks and application-oriented decision-support systems, clarifying the gap it is designed to fill (Section 2.2).

2.1. *CADEMAS framework*

CADEMAS (Novoa-Hernández et al., 2026) provides a general conceptual and architectural framework for structuring cooperation among heterogeneous Automated Decision-Making (ADM) units. Organizational decisions may draw on classifiers, optimization or simulation models, rule-based systems, software agents, or institutional procedures, whose assumptions, data requirements, output types, and reliability may differ substantially. Rather than prescribing a particular learning method or coordination protocol, CADEMAs provides a common structure through which these heterogeneous components can contribute to a shared decision process and their outputs can

subsequently be interpreted in context.

A central premise of the framework is that an ADM unit need not produce a complete operational decision on its own. Instead, it may contribute a *partial decision signal*, such as a score, prediction, rule activation, explanation, or suggested action. The semantics of these signals need not be identical across units, as cooperation does not require the underlying ADM mechanisms to be homogeneous. Their outputs are therefore brought together through an integration layer and subsequently considered in relation to the context in which the decision is made. Accordingly, a CADEMAs instance is represented by the tuple

$$S = \langle \mathcal{IM}, A, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle,$$

where $\mathcal{IM}$ is the Input Manager, A is the set of ADM units, $\mathcal{OM}$ is the Decision Integration Layer, $\mathcal{C}$ is the Contextual Modulator, and $\mathcal{T}$ is the universe of admissible data types and representations.

The Input Manager $\mathcal{IM}$ is responsible for routing, transforming, and validating the information entering the architecture. In particular, it distinguishes training or calibration data, which are used offline to construct or update the internal models of the ADM units, from decision-case data provided when a new case is evaluated. The ADM units are denoted by

$$A = \{ADM_1, \dots, ADM_n\}.$$

Each ADM_k receives a data representation $r_k \in \mathcal{T}$ and produces an output $o_k \in \mathcal{T}$. Depending on the application, cooperation may be *indirect*, with each unit contributing an independent signal, or *direct*, with units exchanging intermediate representations. CADEMAs accommodates both forms while leaving the specific communication protocol to the particular application.

The outputs generated by the ADM units form the heterogeneous collection $O = \{o_1, \dots, o_n\}$. The Decision Integration Layer $\mathcal{OM}$ brings these outputs together and synthesizes them into a unified decision object D through the mapping

$$\Phi : O \rightarrow D.$$

This integration establishes a common representation for the information contributed by the different ADM units and provides the basis for its subsequent consideration in context. The resulting D is a *raw* decision proposal, since the outputs have been harmonized but have not yet been adjusted to the conditions under which the decision is to be used.

The Contextual Modulator then incorporates the relevant contextual information

into this integrated proposal. Formally,

$$\Gamma : D \times \mathcal{C} \rightarrow D, \quad D^{\text{final}} = \Gamma(D^{\text{raw}}, \mathcal{C}),$$

where $\mathcal{C}$ represents the contextual conditions relevant to the decision. These may include regulatory requirements, organizational policies, resource constraints, or other scenario-dependent considerations. Depending on the application, contextual information may be represented through rules, utility functions, constraints, fuzzy propositions, or other complementary mechanisms. The resulting D^{final} therefore represents the decision after the information provided by the ADM units has been integrated and considered in its relevant context.

2.2. *Related frameworks and decision-support applications*

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address how multiple decision entities interact, and application-oriented decision-support systems, which implement concrete methods for a domain or family of decision problems.

At the framework level, cooperative game theory offers one formal perspective on how decision-making entities with heterogeneous roles or attributes may combine their contributions, including settings in which each agent is characterized by multiple attributes rather than a single payoff value (Borkotokey, Hazarika, & Mesiar, 2019). CADEMAs does not adopt this game-theoretic formulation, but shares its concern with making cooperation among heterogeneous entities explicit. At the operational level, Tariq, Chhetri, Nepal, and Paris (2024) propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Related work on hybrid intelligence emphasizes that authority distribution and human interpretive capacity should be modeled separately, because the level of AI integration and the cognitive flexibility of human decision-makers jointly shape decision quality and accountability (Deliu & Moldovan, 2026). A neighboring line of work examines cooperation among multiple AI agents, including LLM-based multi-agent reinforcement learning (Sun, Huang, & Pompili, 2024). These contributions provide relevant points of comparison because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

A related theoretical thread connects cooperative game theory to case-level expla-

nation. The Shapley value distributes a joint outcome among players according to their averaged marginal contributions (Borkotokey et al., 2019) and has been adopted in explainable artificial intelligence as a principled attribution mechanism. SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) treat model features as players and decompose each prediction into additive feature contributions with axiomatic guarantees; recent surveys document the breadth of this cooperative-to-XAI bridge (Li, Sun, Huang, & Chen, 2024). This literature motivates the inclusion of an explicit explanation module in CADEMAs-ML. The current implementation does not compute Shapley values over features: because the uploaded H2O MOJO models are treated as black boxes and may include stacked ensembles, the tool uses a model-agnostic perturbation procedure instead. The purpose is nonetheless the same as in Shapley-inspired XAI: to make the local drivers of a recommendation inspectable alongside the cooperative and contextual scores.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION (Więckowski & Sałabun, 2024) and the fuzzy DEMATEL-based tool of Chekry et al. (2024) emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored hybrid pipelines that combine expert judgment, causal weighting, and predictive components in domain-specific settings (Abdulgader, Eid, & Daneshvar Rouyendegh, 2018; Özkurt, Canay, Tunç, Aydın, & Velioğlu, 2025; Sahraei et al., 2022). Recent work has also extended the fuzzy toolbox through aggregation operators that account for interactions among decision attributes, capturing dependency structures that simple weighted averages cannot represent (Kacprzyk, Sirbiladze, & Tsulaia, 2022). The importance of explainability in AI-based decision support has been broadly recognized: any automated system whose outputs must be justified or audited requires transparent mechanisms to convey its reasoning to stakeholders (Barredo Arrieta et al., 2020), a requirement increasingly addressed through XAI techniques, including Shapley-value attribution (Li et al., 2024; Lundberg & Lee, 2017), in hybrid ranking and prediction pipelines (Özkurt et al., 2025). These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs decomposition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including PYFDM (Więckowski, Kizielewicz, & Sałabun, 2022), PYIFDM (Więckowski, Kizielewicz, & Sałabun, 2023), the IFMCDM package of Roszkowska, Jefmański, Dudek, and Kusterka-Jefmańska (2024), the ELIGERE platform (Grazioso, Caporaso, & di Gironimo, 2023), and the intuitionistic fuzzy dimensional-analysis approach of Pérez-Domínguez, Callejas-Cuervo, and García-Luna (2024), are also relevant. They improve reproducibility and provide reusable computational building blocks.

Table 1. Comparison of representative application-oriented decision-support works related to CADEMAs-ML.

Work	Decision-support focus	Multiple ADM units	Explicit context layer	Case-level XAI	Relation to CADEMAs-ML
Tariq et al. (2024)	Human–AI collaborative decision framework with automated, augmented, and collaborative modes	Partly	Partly	No	Strong framework-level precedent, but not a configurable fuzzy/ML prioritization workflow
Więckowski and Sałabun (2024)	Auditable fuzzy MCDM through a web-based tool	No	Partly	No	Strong transparency, but no cooperative ADM integration layer
Chekry et al. (2024)	Open-source fuzzy DEMATEL decision-support tool	No	Partly	No	Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like
Özkurt et al. (2025)	Hybrid fuzzy MCDM, ML, and XAI for transparent ranking and prediction	Partly	Partly	Partly	Combines ranking, prediction, and XAI, but lacks a general cooperative ADM architecture
Sahraei et al. (2022)	Fuzzy DEMATEL/BWM/ANP pipeline for spatial risk prioritization	No	Partly	No	Explicit causal weighting under uncertainty, but application-specific
Godz et al. (2026)	Employee-attribution prevention using CADEMAs with departmental ML models	Yes	Yes	No	Direct CADEMAs application that motivates the implemented workflow
Novoa-Hernández et al. (2026)	Aggregation semantics for CADEMAs prioritization under alternative operators	Yes	Yes	No	Analyzes how linear, geometric, and minimum rules preserve or compensate contextual requirements
Vazquez-Sánchez et al. (2026)	Cooperative ADM with fuzzy contextual modulation for SME insolvency prediction	Yes	Yes	Yes	Extends CADEMAs-style cooperation to survival models with contextual modulation and XAI support
CADEMAs-ML (this work)	Configurable prioritization with ML-based ADM units, fuzzy context scenarios, and local case explanation	Yes	Yes	Yes	Operationalizes CADEMAs for reusable, explainable ML-driven decision support

Table 1 summarizes representative application-oriented contributions in these areas. Three observations emerge from this comparison. First, published systems already provide strong ingredients for transparent decision support, including human–AI collaboration, auditable fuzzy MCDM, hybrid ranking pipelines, and reusable software libraries (Özkurt et al., 2025; Tariq et al., 2024; Więckowski et al., 2022; Więckowski & Sałabun, 2024). Second, these ingredients are usually developed in isolation. Third,

recent CADEMAs applications already combine cooperative multi-ADM integration with an explicit contextual layer (Godz et al., 2026; Novoa-Hernández et al., 2026; Vazquez-Sánchez et al., 2026), yet they remain domain-specific configurations rather than a reusable software workflow with case-level explanation. What remains missing is an operational implementation that keeps these capabilities configurable and auditable together.

CADEMAS-ML is designed to address this gap for the predictive-model setting. Building on the framework and its attrition instantiation discussed above, it provides a configurable software implementation in which ML-based ADM outputs, contextual reasoning, and case-level explanation coexist as explicit and auditable components of the decision-making process. The source code is publicly available under an MIT license; the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

References

- Abdulgader, F. S., Eid, R., & Daneshvar Rouyendegh, B. (2018, November). Development of decision support model for selecting a maintenance plan using a fuzzy MCDM approach: A theoretical framework. *Applied Computational Intelligence and Soft Computing*, 2018, 1–14.
- Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., . . . Herrera, F. (2020, June). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.
- Borkotokey, S., Hazarika, P., & Mesiar, R. (2019, September). Cooperative games with multiple attributes. *International Journal of General Systems*, 48(8), 825–842.
- Bui, T., & Lee, J. (1999). An agent-based framework for building decision support systems. *Decision Support Systems*, 25(3), 225–237.
- Chekry, A., Bakkas, J., Hanine, M., Montero, E. C., Garat de Marin, M. S., & Ashraf, I. (2024). PyDEMATEL: A Python-based tool implementing DEMATEL and fuzzy DEMATEL methods for improved decision making. *SoftwareX*.
- Deliu, D., & Moldovan, A.-L. (2026). Human–AI collaboration and cognitive flexibility in industry 6.0: A framework for purposeful decision-making in innovation and project management. *International Journal of Managing Projects in Business*.
- Godz, M., Novoa-Hernández, P., & Pelta, D. A. (2026). A cooperative and context-aware approach for employee attrition prevention. In *Information processing and management of uncertainty in knowledge-based systems* (pp. 203–217). Springer Nature Switzerland.
- Grazioso, S., Caporaso, T., & di Gironimo, G. (2023). Decision support in engineering design: the ELIGERE open source software platform. *International Journal on Interactive Design and Manufacturing (IJIDeM)*.
- Herrera-Viedma, E., Herrera, F., & Chiclana, F. (2002). A consensus model for multiperson decision making with different preference structures. *IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans*, 32(3), 394–402.

- Kacprzyk, J., Sirbiladze, G., & Tsulaia, G. (2022). Associated fuzzy probabilities in MADM with interacting attributes: Application in multi-objective facility location selection problem. *International Journal of Information Technology & Decision Making*, 21(4), 1155–1188.
- Lamata, M. T., Pelta, D. A., & Verdegay, J. L. (2020, oct). The role of the context in decision and optimization problems. In *Fuzzy approaches for soft computing and approximate reasoning: Theories and applications* (pp. 75–84). Springer International Publishing.
- Li, M., Sun, H., Huang, Y., & Chen, H. (2024). Shapley value: From cooperative game to explainable artificial intelligence. *Autonomous Intelligent Systems*, 4(1), 2.
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., ... Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in neural information processing systems* (Vol. 30, pp. 4765–4774). Curran Associates. Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- Navarro-Galera, A., Lara-Rubio, J., Novoa-Hernández, P., & Cruz Corona, C. A. (2025, March). Using decision trees to predict insolvency in spanish smes: Is early warning possible? *Computational Economics*, 65(1), 91–116. Retrieved from <http://dx.doi.org/10.1007/s10614-024-10586-5>
- Novoa-Hernández, P., Godz, M., & Pelta, D. A. (2026). Aggregation semantics for prioritization in cooperative automated decision-making. *Mathematics*, 14(18). Retrieved from <https://www.mdpi.com/2227-7390/14/18/3397>
- Novoa-Hernández, P., Pelta, D. A., Godz, M., Verdegay, J. L., & Buendia-Carrillo, D. (2026). Cademas – a framework for cooperative automated decision-making systems. In *2026 ieee conference on artificial intelligence (cai)* (p. 756–761).
- Özkurt, C., Canay, O., Tunç, E. A., Aydin, E., & Velioglu, B. S. (2025). Renewable energy source ranking and analysis using fuzzy MCDM, ML, and XAI techniques. *Baltic Journal of Modern Computing*, 13(3), 656–679.
- Pérez-Cañedo, B., Porras, C., Pelta, D. A., & Verdegay, J. L. (2023, may). Modeling contexts as fuzzy propositions in optimization problems. *IEEE Transactions on Fuzzy Systems*, 31(5), 1474–1483.
- Pérez-Domínguez, L., Callejas-Cuervo, M., & García-Luna, F. (2024). Intuitionistic fuzzy dimensional analysis in multi-criteria decision-making: A computational approach. *IEEE Access*, 12, 1–15.
- Roszkowska, E., Jefmański, B., Dudek, A., & Kusterka-Jefmańska, M. (2024). IFMCDM: An R package for intuitionistic fuzzy multi-criteria decision making methods. *SoftwareX*, 26, 101676.
- Sahraei, R., Kanani-Sadat, Y., Homayouni, S., Safari, A., Oubennaceur, K., & Chokmani, K. (2022). A novel hybrid GIS-based multi-criteria decision-making approach for flood susceptibility analysis in large ungauged watersheds. *Journal of Flood Risk Management*, 16(2), e12871.
- Salo, A. A. (1995). Interactive decision aiding for group decision support. *European Journal*

- of Operational Research*, 84(1), 134–149.
- Sun, C., Huang, S., & Pompili, D. (2024). LLM-based multi-agent reinforcement learning: Current and future directions. *AI*, 5(3), 1234–1252.
- Tariq, S., Chhetri, M. B., Nepal, S., & Paris, C. (2024). A2C: A modular multi-stage collaborative decision framework for human-AI teams. *Expert Systems with Applications*, 238, 122334.
- Vahidov, R., & Fazlollahi, B. (2004). Pluralistic multi-agent decision support system: A framework and an empirical test. *Information & Management*, 41(7), 883–898.
- Vazquez-Sánchez, A. A., Buendía-Carrillo, D., & Corona, C. C. (2026). A cooperative automated decision-making strategy with fuzzy contextual modulation for SME insolvency prediction. *Mathematics*, 14(19). Retrieved from <https://www.mdpi.com/2227-7390/14/19/3516>
- Więckowski, J., Kizielewicz, B., & Sałabun, W. (2022, December). pyFDM: A Python library for uncertainty decision analysis methods. *SoftwareX*, 20, 101271.
- Więckowski, J., Kizielewicz, B., & Sałabun, W. (2023, May). Handling decision-making in intuitionistic fuzzy environment: PyIFDM package. *SoftwareX*, 22, 101344.
- Więckowski, J., & Sałabun, W. (2024). MakeDecision: Online system for the graphical design of decision-making models in crisp and fuzzy environments. *SoftwareX*.